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	The laws of physics (referred to Einstein's equivalence principle) dictate that any linear accelerometer will measure the same acceleration independently of whether it results from translation or the action of a gravitational field. Indeed, primary afferent neurons innervating otolith receptors in the vertebrate ear only detect the total gravitoinertial acceleration and produce equivalent responses to both tilts relative to gravity and to translational movements. Yet, the central vestibular system may correctly discriminate the acceleration source in order to generate appropriate reflexes and positively contribute to spatial orientation and spatial memory of extra-personal space. How the brain solves the tilt/translation ambiguity of otolith afferent information has been a matter of debate for many decades. One of the most influential ideas is based on what is referred to as the "multisensory integration" hypothesis, which states that otolith afferent information alone is not sufficient to correctly discriminate between head translation and tilt. Instead, to encode		

Task Description:

motion, the brain relies on signals from multiple sensors (e.g., signals from both the semicircular canal and the otolith organs). Another proposal which has become popular in the past decade is referred to as the "frequency segregation" hypothesis. It states that the discrimination between gravity and translational accelerations could be performed by utilizing otolith signals alone, if it is assumed that the functional role for each component of gravitoinertial acceleration is segregated in frequency. Accordingly, high frequency accelerations are interpreted as translations whereas low frequency accelerations are interpreted as tilts. Whereas the two hypotheses are not necessarily mutually exclusive, the "multisensory integration" hypothesis has received recent support through several theoretical and experimental studies. Specifically, recent work in our laboratory has clearly demonstrated that signals from the semicircular canals provide necessary and critical sensory cues to detect correctly the translational component of motion, at least during mid and high frequency head movements (frequencies > 0.1 Hz). Without functional signals from the semicircular canals (e.g., in canal-plugged animals), the translational VOR (TVOR) is proportional to the total gravitoinertial acceleration, rather than the true translational component of the movement. The processing and detection of gravity and translational accelerations represent some of the most fundamental issues for understanding vestibular sensorimotor control in normal gravity, as well as in altered gravity environments.
The present studies propose to extend this work with the goal of understanding the underlying multisensory interactions governing not only the VORs but also our spatial memory as it is continuously updated to direct voluntary eye, head and arm movements. Specifically, the following objectives are proposed:

Specific Aim 1: To investigate the temporal aspects of tilt/translation discrimination in the VOR and develop/modify/test a dynamic model of otolith/canal interactions to process gravitoinertial accelerations in the generation of the VORs. Our previous experiments used steady-state sinusoidal stimulation in a limited frequency range (0.16 ? 1 Hz). Preliminary data, however, suggest that the dynamic interactions between otolith and semicircular canal signals is rather complex and non-linear. To acquire further insight into the dynamic aspects of this sensorimotor processing, we will use transient stimuli of tilt/translation combinations. Differences in latency, initial eye acceleration and velocity of the TVOR as a function of vergence angle will be used to expand and modify existing non-linear models of otolith/canal interactions, as well as incorporate these interactions within newly developed models of shared dynamic processing in the RVOR/TVOR.

Specific Aim 2: To investigate the role of extra-vestibular signals in the process of neural discrimination between tilt and translation in more natural situations with the head free. Our previous experiments were conducted in head-fixed animals in darkness, where the only available sensory information that could be used to detect motion arises in the peripheral vestibular system. However, multiple sensory signals are activated during natural movements. As a first step to start addressing the role of extra-vestibular signals in the brain?s ability to discriminate tilt from translation, we will test the animals with the head free. As we have previously shown, head-fixed animals with inactivated semicircular canals fail to correctly discriminate linear translations from tilts relative to gravity. This is the case even after several months following the plugging operation (i.e., in head-fixed animals, extra-vestibular signals can not substitute for the missing canal function). We hypothesize, however, that extra-vestibular (e.g., proprioceptive and efference copy) signals activated when the head is free can contribute to this task, and that this contribution is particularly evident after sensorimotor adaptation in canal-plugged animals.

Specific Aim 3: To investigate the role of otolith and extra-otolith signals in updating spatial memory for sensorimotor tasks during voluntary motor behavior (vestibular memory-contingent saccades and reaching arm movements). The functional utility of the vestibular system is not only limited to reflexes. An important role of the vestibular system is to update our spatial memory of the extra-personal space. Whether such ?higher level? vestibular processing mechanisms utilize that same rules and computations for the detection and discrimination of gravity as those demonstrated in brainstem reflexes (e.g., the TVOR) is unknown. We propose to investigate the role of otolith and extra-otolith (i.e., semicircular canal) signals in the processing of gravitoinertial accelerations for spatial memory, as accessed during voluntary motor behavior (vestibular memory-contingent saccades and reaching). These experiments will be conducted in both animals with intact labyrinths and animals with all semicircular canals inactivated by plugging. Specifically, we will test the hypotheses that (a) gravitoinertial signals are appropriately interpreted in labyrinthine-intact animals and are used to accurately update spatial memory for the generation of appropriate memory-guided behavior (saccades and reaching) and (b) the computational principles and rules used by the brain to achieve these voluntary sensorimotor tasks are similar to those used in the TVOR, suggesting a common processing.

Specific Aim 4: To investigate the role of the vestibulo-cerebellum in the sensorimotor task of distinguishing between translational accelerations and gravity. It is well known that the cerebellar nodulus and ventral uvula have direct reciprocal connections with the vestibular nuclei and receive direct input from vestibular afferents. In addition, surgical lesions of the cerebellar nodulus and ventral uvula have been previously shown to compromise the dynamic properties and 3-D organization of the inertial (velocity storage) system. Thus, it is hypothesized that intact and functional cerebellar nodulus and ventral uvula are also important for the necessary otolith/semicircular canal interactions needed to functionally segregate gravitoinertial accelerations into gravity and translational components. To investigate this hypothesis, tilt/translation discrimination protocols (including both the VOR and vestibular memory-contingent saccades and reaching) will be quantitatively compared before and after surgical lesions of the nodulus and ventral uvula. It is hypothesized that lesions of the nodulus/ventral uvula will result in significant deficits in the sensorimotor task of tilt/translation discrimination, as reflected not only in the VOR but also in accurate vestibular memory-contingent voluntary motor behavior.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Throughout the history of the manned space flight program, microgravity has produced vestibular-related symptoms that result in personal discomfort and a loss in crew performance. The symptoms subside within several days of microgravity exposure, suggesting that interactive multisensory mechanisms may be responsible for the initiation of symptoms and the subsequent adaptation. Specifically, the processing and detection of gravity and translational accelerations is one of the most fundamental issues for understanding vestibular sensorimotor control in normal gravity, as well as in altered gravity environments. In order to better understand the nature of multisensory signals in detecting gravity, we will study the different sensory system contributions in normal, canal-plugged and cerebellar-lesioned non-human primates. The present experiments address the relative roles of semicircular canals and otolith information, as well as extra-vestibular (i.e., proprioceptive and efference copy) signals in the detection and discrimination between translational and tilt motion during both reflexive and voluntary motor actions. The proposed studies focus on four important aspects: the dynamics of the semicircular canal/otolith interactions (Aim 1), the role of proprioceptive and efference copy signals during

motion in head-free animals (Aim 2), the role of tilt/translation discrimination in spatial memory and memory-guided voluntary motor control (e.g., saccades and reaching; Aim 3), as well as the potential role of the vestibulo-cerebellum in these multisensory interactions (Aim 4). It is hypothesized that correct tilt/translation discrimination and spatial memory update are ultimately linked to an integration of multi-sensory information, where the cerebellum (and particularly the posterior cerebellar vermis) is critical for proper function. Unless these multi-sensory interactions are better understood in normal gravity, our understanding of neurovestibular adaptation to altered gravity and the development of relevant countermeasures remains largely incomplete.

Specific aim 1 of the original grant aimed at characterizing the dynamics and viewing distance-dependence of the horizontal eye movements elicited during transient lateral translation, transient roll tilt or combinations of the two. A second goal was to use the experimental results to formulate a physiologically-realistic model of otolith/canal convergence to detect inertial head movements. We have completed this aim, with the following main results. It has been traditionally believed that semicircular canal signals drive compensatory responses to head rotations (Rotational VOR), whereas otolith signals compensate for translational movements (Translational VOR). However, a sensory ambiguity exists because otolith afferents are activated similarly during head translations and reorientations relative to gravity (i.e., tilts). How then are otolith signals that are appropriately routed to generate the TVOR during pure head translations “eliminated" from driving this reflex during mid and high frequency roll head movements? Previous work has suggested that rotation signals from the semicircular canals could provide the extra-otolith cues necessary to ensure that dynamic head tilts do not elicit a TVOR. Although this problem is fundamental for gaze stabilization, a quantitative solution that is physiologically plausible (i.e., not based simply on abstract nonlinear vector algebra) has so far been missing. Thus, we sought to investigate how extra-otolith signals contribute. For this, novel stimulus combinations were used to quantitatively characterize, for the first time, the temporal and viewing-distance-dependent properties of a transient TVOR elicited in the absence of a lateral acceleration stimulus to the otoliths during combined translational/rotational motion. Experimental evidence has been gathered showing that integrated angular velocity signals from the semicircular canals combine centrally with otolith information in order to elicit functionally appropriate VORs. Such integrated canal signals, necessary to “eliminate” the otolith drive to horizontal eye movements during head reorientations relative to gravity, are salient but typically hidden during pure translational head rotation. Yet, they become unmasked and appear as a canal-driven TVOR during combined translation and rotation resulting in the absence of an interaural otolith stimulus. This canal-driven response to translation exhibits the appropriate dynamic properties and scaling with viewing distance required to provide a fully functional complement to the otolith-driven TVOR. These experimental results have been complemented with the development of a model, restricted to horizontal eye movements for simplicity, that can be simulated to predict the basic characteristics of our experimental results. The model delineates a potential neural pathway for the canal-driven component of the TVOR by incorporating an additional neural filter, that we postulate could be the so-called “velocity storage integrator”. Accordingly, we proposed a novel functional purpose for the integrative network known as “velocity storage” in high-frequency gaze stabilization (rather than low frequency rotational responses, as typically considered). These results have been published (Green and Angelaki 2003). These results have also motivated the development of an expanded model of tilt/translation discrimination that is consistent with the known anatomy and neurophysiology and which can predict the observed VN neuron responses. This model has also been published (Green and Angelaki 2004).
Specific Aim 2 of the previous grant was to compare tilt/translation discrimination in normal and canal-plugged animals in situations of active head movements with the head free. However, it quickly became apparent that two control studies had to be performed prior to the proposed experiments. First, we showed that the otolith-driven eye movements in canal-plugged animals adaptively change over time (Angelaki, Newlands and Dickman 2002). Thus, it became clear that our proposed studies needed to be performed in the acute stage, prior to these adaptive changes. Second, there is strong neurophysiological evidence, as well as some behavioral support, that a vestibulocollic reflex, which includes a compensatory head rotation, might be part of the gaze stabilization mechanism during translation in head-free subjects. Before we could proceed with aim 2, we decided to investigate this hypothesis. Thus, we recorded eye and head movements in four rhesus monkeys whose head was free to rotate in the horizontal plane. We found no evidence for a functionally-relevant head horizontal rotation component during passive lateral translational motion (Wei and Angelaki 2004). As both of these ‘control’ studies are now nearly complete, we plan to continue with the original aim 2, in the near future.

Specific Aim 3 of the grant proposed to study tilt/translation discrimination in spatial memory and memory-guided voluntary motor control (e.g., memory-guided saccades). We first trained three animals to make memory-guided saccades when placed at different head orientations in space. We were intrigued by the up/down asymmetry of memory saccades and we wanted to know if it depended on head orientation relative to gravity. Thus, we collected memory-guided saccades in different head orientations in space. We found that the up/down asymmetry previously known to characterize memory-guided saccades is expressed in head (or body), rather than space (gravity) coordinates. This work has been published (Vogelstein et al. 2003). We also studied how smooth pursuit changes as a function of static head orientation relative to gravity, as well as the accuracy of the TVOR during forward/backward heading. This work is also published (Hess and Angelaki 2003a,b). We are currently training two of these animals to make vestibular-memory-contingent saccades, with the goal to perform the outlined experiments soon.

Neurophysiological studies supported by this grant: In preparation for a competitive renewal of this grant application, we started recording from single neurons in the vestibular nuclei in order to understand how central vestibular neurons process gravity and translational accelerations. First, we performed a survey of the otolith/canal convergent properties in the VN. This resulted in two publications, one characterizing the 3D properties (Dickman and Angelaki 2002) and the other characterizing their response dynamics (Dickman and Angelaki 2004). To provide a comprehensive comparison with central responses, we also characterized both the dynamics and spatial sensitivity vectors of afferent response properties using identical stimuli and data analyses (Haque, Angelaki and Dickman 2004). We have also extended these studies to the most medial of the deep cerebellar nuclei, the rostral fastigial nucleus. The first manuscript on the properties of these neurons has been published (Shaikh, Meng and Angelaki 2004). Finally, we recently used a combined tilt/translation protocol to characterize if central neurons encode the translational or total gravitoinertial acceleration. This work has been accepted in Nature (Angelaki, Shaikh, Green and Dickman, in press).

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